

LEVEL 26 STRUCTURAL VERIFICATION — TATE, MAZUR, RIBET, KOLYVAGIN HONEST DISPLAYS IN LEAN 4.12

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ABSTRACT. The Beal equation $A^x + B^y = C^z$ with $x, y, z > 2$ is studied here only for the restricted one-parameter family $C = B + 3$ inside the signature $(4, 4, 13)$, i.e. $A^4 + B^4 = (B + 3)^{13}$. This “gap-3” family is an internal naming convention of a larger, still largely open Lean development; it is *not* derived in this repository from the unrestricted equation $A^4 + B^4 = C^{13}$, and no claim about the general signature is made. For the single specialization $(A, B) = (29, 1)$ the associated Frey curve $E : y^2 = x(x - A^4)(x + B^4)$ has displayed conductor $928 = 2^5 \cdot 29$, Néron types I_0^* at 2 and I_8 at 29, and valuations $v_2(\Delta) = 6$, $v_{29}(\Delta) = 8$; Tate’s algorithm itself is not formalized in Mathlib 4.12, and the same family at $(A, B) = (1, 1)$ displays conductor 32, not 928. This draft records the Lean 4.12 verification of four families of displayed numeral and Finset facts, named only by association with Tate, Mazur, Ribet, and Kolyvagin; none of the four classical theorems is proved. Mazur-named display: $|\mathrm{SL}_2(\mathbb{F}_{13})| = 2184$, $48 < 2184$, $13 = 2^2 + 3^2$, $288/48 = 6$, cusp Finset card 2; $X_0(13)$ has genus 0, so $X_0(13)(\mathbb{Q})$ is infinite and no 13-isogeny is excluded by this display. Ribet-named display: $928/29 = 32$, an empty matching Finset (card 0) against the LMFDB dimension-1 newform space at level 32 (32a1). Kolyvagin-named display: $|\mathrm{Sel}_2| = 1$, $3 \cdot 7 = 21$, $L/\Omega = 1/3$ and $1/7$ on Cremona 26a1/26b1, and the typeclass upgrade $\mathrm{Nonempty}(\mathrm{Fintype} \alpha) \rightarrow \mathrm{Finite} \alpha$; the parent identifier `IsRankZero := Subsingleton` is mathematically false on both curves, which have nontrivial torsion. Separately, the repository proves a genuine finite lemma, `no_match_32a1_of_good_red_29`: on the gap-3 equation with $29 \nmid A$, $29 \nmid B$, $29 \nmid B + 3$, the Frey trace at 29 cannot equal -10 , the trace of 32a1, by an exhaustive residue check compiled with `native_decide` (axiom `Lean.ofReduceBool`). This lemma is not connected in Lean to Ribet’s level-lowering step, which stays an uninhabited `def Prop`, and it does not address $29 \mid ABC$. The four named displays themselves use explicit types and `decide`, not `native_decide`. Version DOI 10.5281/zenodo.22863527; concept DOI 10.5281/zenodo.22379293.

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1. INTRODUCTION

The Beal conjecture asserts that if $A^x + B^y = C^z$ with integers $A, B, C \neq 0$ and exponents $x, y, z > 2$, then A, B, C share a prime factor [7]. This draft does not prove the conjecture, does not address the unrestricted signature $(4, 4, 13)$ equation

$$A^4 + B^4 = C^{13},$$

and does not derive any reduction of that equation. It records a *structural verification* of Lean 4.12 material attached to a single, deliberately restricted one-parameter family inside that signature, together with an honest account of what each Lean name does and does not establish.

The gap-3 family: what it is and where it comes from. Throughout, “the gap-3 equation” means

$$A^4 + B^4 = (B + 3)^{13}, \tag{1}$$

i.e. $C := B + 3$. This is *not* a reduction proved from the unrestricted equation anywhere in the repositories accompanying this draft. It is the name given, inside a much larger and mostly uninhabited Lean development (files `BealGap3BakerUpperBound.lean`, `BealFreyB14.lean`, `DarmonMerelFrey4413.lean`, and the `Beal_4_13_13_Zsigmondy_13_Step*` series), to one specific test family that the author chose to formalize as a vehicle for exercising Tate/Mazur/Ribet/Kolyvagin-style arguments in Lean. Four points, requested explicitly by an external reviewer, are recorded here rather than left implicit:

- (1) *Where $C = B + 3$ comes from.* It is an adopted normalization, not a derived one. The surrounding Lean files contain a size lemma for the *different* triple $(4, 13, 13)$ showing $C^{13} - B^{13} \geq 26B^{12}$ when $C \geq B + 2$ (`Beal_4_13_13_Size_C_ge_B_plus_2.lean`) and an observation that $\gcd(B, B + 3) \mid 3$ (`gcd_B_Bplus3_dvd_three` in `BealFreyB14.lean`), but neither of these, nor anything else found in the repository, is a theorem that a hypothetical $(4, 4, 13)$ counterexample must have $C - B = 3$. The gap-3 family is a case study, not a WLOG reduction.
- (2) *What it covers.* Only the one-parameter slice (1), for arbitrary $A, B \in \mathbb{N}$. It says nothing about $C - B \in \{1, 2, 4, 5, \dots\}$ or about solutions, if any, of the unrestricted equation.
- (3) *Whether the “good reduction at 29” hypotheses hold in any intended case.* They do not hold automatically; they are hypotheses of the lemma discussed in Section 4, and no Lean theorem in the repository shows that a hypothetical solution of (1) must satisfy $29 \nmid ABC$. The case $29 \mid ABC$ is untouched.
- (4) *Whether a trace mismatch eliminates the level-32 newform after level lowering.* It does not, in Lean, because the level-lowering step itself (`ribet_lowers_Frey_to_level_32`) stays an uninhabited `def Prop`: there is no formal link between the actual Galois representation of the Frey curve and the newform `32a1` whose trace is being compared.

Section 4 returns to this with the specific lemma statement, its hypotheses, and its axiom profile.

The strategy, honestly labelled. The classical strategy this development is modeled on is Frey–Serre–Ribet–Mazur [4, 10, 9, 8]: compute the conductor of a Frey curve by Tate’s algorithm [11], exclude a rational 13-isogeny by Mazur’s theorem on $X_0(13)$, lower the residual level in the style of Ribet, and compare the resulting newform space with Mordell–Weil data via the Cremona labels `26a1` and `26b1` [1, 6]. None of those four classical theorems is proved, used, or even stated as a

hypothesis-free implication anywhere in the Lean files discussed below. The Frey curve of (1) is the displayed Weierstrass model

$$y^2 = x(x - A^4)(x + B^4) \quad \text{i.e.} \quad [0, B^4 - A^4, 0, -A^4B^4, 0]. \quad (2)$$

The short model $[0, -(A^4 + B^4), 0, 0, 0]$ is a different curve and is not used as E . Sections 2–5 are titled “displayed data” for exactly this reason: each records what four Lean files, named *after* the classical authors, actually contain — numerals, Finset cardinalities, and one typeclass upgrade — while the corresponding algebraic statements of Tate/Mazur/Ribet/Kolyvagin remain uninhabited `def Prop` in the parent files.

Pins and provenance. The formal pin is Lean v4.12.0, mathlib commit 809c3fb3b5c8f5d7dace56e200b426187516535a with inputRev v4.12.0 (not @ main), and the beal-conjecture Lake manifest prefix 83739542. The command `lake update` was not run. The vendor tree `Level26/BealLevel26Foundations` at db7a556/fea0c393 has difference 0 against b780c9c/1a00fbd. Version DOI 10.5281/zenodo.22863527 is a new version of 22854603 and 22851334; the concept DOI 10.5281/zenodo.22379293 stays, related as `isVersionOf` [3, 2].

This is a Mathematics of Computation manuscript draft, not an arXiv preprint. There is no arXiv identifier and no arXiv badge.

The four display files live in `Level26/HonestB0Search/` on `beal-conjecture main` (release tag `v30.0.0-level-26-structural-verification`, commit `af520d3`) and on `foundations phase-darmon-merel-4413` at `2b60ccd`. At this writing the `HonestB0Search` slices are near identical; `beal-conjecture` is the superset, carrying the historical `BealMatveevBeal` target in addition to the four finals. Any commit hash for `main` cited elsewhere in this draft for purposes other than pinning a specific historical file is only accurate as of the revision that introduced it; the release tag above is the stable anchor.

2. DISPLAYED DATA I — A TATE-STYLE CONDUCTOR NUMERAL AT $(A, B) = (29, 1)$

The file `Tate_Frey_Conductor_29_Neron_final.lean` (11c0ad7; foundations incremental 35488239286, 137s cache hit) packages the displayed conductor numeral, for the single specialization $(A, B) = (29, 1)$ of the gap-3 equation (1),

$$N_E^{\text{disp}} = (2 : \mathbb{N})^5 \cdot 29 = 928.$$

The named theorems `displayed_Neron_conductor` and `final_N_E_eq_928` are that equality, proved by `decide`. The split `displayed_928_div_29` is $928/29 = 32$. At the other named specialization $(A, B) = (1, 1)$ the same family displays conductor 32, not 928 (`sagemath/certs/tate_nero_29.json`): the numeral 928 is a fact about one pair (A, B) , not a property of every solution of (1).

Display 2.1 (Displayed Néron conductor). In Lean 4.12, with `open Nat Finset Classical`:

- $(2 : \mathbb{N})^5 \cdot 29 = 928$ (`final_N_E_eq_928`);
- $928/29 = 32$ (`displayed_928_div_29`);
- $32 \cdot 29 = 928$;
- $-29 \mid 32$.

These are numerals. Mathlib 4.12 has no `KodairaType` and no `NeronModel`. The name $N_E = 928$ is not the output of a Lean formalization of Tate’s algorithm or of a Néron model.

The displayed Weierstrass invariants of (2) are

$$c_4 = 16(A^8 + B^8 + A^4B^4), \quad \Delta = 16A^8B^8(A^4 + B^4)^2.$$

On the inhabited slice one has $v_2(\Delta(1, 1)) = 6$ (type I_0^* valuation) and $v_{29}(\Delta(29, 1)) = 8$ (type I_8), together with $29 \nmid c_4(29, 1)$. The residual-level Finset $\{32, 928\}$ has card 2; that card is a Finset numeral over two named specializations, not two Néron models of a single curve. The parent names

`Tate_algorithm_at_29`, `Frey_Neron_conductor`, and `Frey_conductor_29_is_Neron` are *theorems in this namespace*: each re-exports the inhabited conjunction `Tate_algorithm_at_2_and_29`. On the parent file `Tate_Frey_Conductor_29.lean` those three names stay `def Prop`.

Honesty note 2.2. The phrase “final $N_E = 928$ ” means the displayed identity $2^5 \cdot 29 = 928$ at $(A, B) = (29, 1)$. It is not Tate’s algorithm as a function on integral models, and it is not a claim about the conductor for other (A, B) satisfying (1). SAGE `sagemath/tate_nero_29.sage` uses the displayed Frey (2), not the short model, and its own comment records that $(1, 1)$ gives 32.

Axioms on the packaged Tate final are `[propext]` for the pure numerals and `[propext, Classical.choice, Quot.sound]` for the conjunction that mentions valuations already inhabited upstream. There is no `Lean.ofReduceBool`, no `sorryAx`, and no new axiom.

3. DISPLAYED DATA II — MAZUR-STYLE NUMERALS FOR $X_0(13)$

The file `Mazur_X0_13_No_Isogeny_final.lean` (2f4e16e; incremental 35488699511, 252s) displays numerals associated with $X_0(13)$ [8]. None of them excludes a rational 13-isogeny; see the honesty note below.

Display 3.1 (Mazur numerals). • $|\mathrm{SL}_2(\mathbb{F}_{13})| = 13 \cdot 12 \cdot 14 = 2184$ (`SL2_F13_card`);

- $48 < 2184$ (`forty_eight_lt_2184`);
- $13 = 2^2 + 3^2$ (`thirteen_eq_two_squares`);
- $288/48 = 6$ (`two_eighty_eight_div_forty_eight`);
- $(\{0, 1\} : \mathrm{Finset} \mathbb{N}).\mathrm{card} = 2$ (`displayed_cusps_card`).

The first four identities are axiom-free `decide` theorems. The cusp-card theorem uses `[propext, Quot.sound]`.

The parent names `X0_13_Q_infinite`, `frey_no_rational_13_isogeny`, `Serre_non_Borel_mod_13`, and `mazur_no_Frey_13_isogeny` are theorems in this namespace. Each re-exports the inhabited conjunction of Display 3.1 with the genus `Nat 0`; despite the names, none of them states or proves that a rational 13-isogeny is excluded. On the parent file `Mazur_X0_13_No_Isogeny.lean` those four names stay `def Prop`, and it is that parent `Prop`, not the theorem in this file, that would correspond to an actual exclusion statement.

Honesty note 3.2 (Genus 0 versus two cusps, and no isogeny exclusion). $X_0(13)$ has genus 0, so $X_0(13)(\mathbb{Q})$ is infinite over $\mathbb{Q}$. The pasted equality $\{2 \text{ cusps}\} = X_0(13)(\mathbb{Q})$ is literature-false and is *not* a theorem. The honest Lean display is finite card 2 versus the genus-0 note `X0_13_genus_nat ≠ X0_13_cusp_count`. The theorem named `X0_13_Q_infinite` packages that conjunction; it is not a Mathlib proof that the rational points are infinite, and it is not Mazur’s isogeny theorem. In particular, the introduction’s classical strategy paragraph names Mazur’s theorem for context only: no Lean theorem in this repository excludes a rational 13-isogeny of the Frey curve, and the genus-0/infinity fact recorded here is logically unrelated to such an exclusion.

Geometrically, a genus-0 curve has trivial Jacobian, so $J_0(13)$ is a point (rank 0, finite). That remark is not claimed as a Lean Mordell–Weil theorem in the display file. Mazur’s theorem that the only rational 13-isogenies of elliptic curves over $\mathbb{Q}$ arise from the cusps remains `def Prop` on the parent, uninhabited.

Axioms: numerals `[propext]` or axiom-free; package `[propext, Quot.sound]`; LLL no-go persistence `[propext, Classical.choice, Quot.sound]`. No `native_decide`, no `sorryAx`, no new axiom.

4. DISPLAYED DATA III — RIBET-STYLE NUMERALS FOR LEVEL 32, AND A CONDITIONAL TRACE LEMMA AT 29

The packaged display. The file `Ribet_Level_Lowering_29_to_32_final.lean` (5631fd1; incremental 35489274297, 282s) displays the residual-level arithmetic named after Ribet [9] without inhabiting abstract level lowering.

Display 4.1 (Ribet numerals). • $928/29 = 32$, $32 \cdot 29 = 928$, $-29 \mid 32$;

- $2^4 = 16$, $16 \mid 32$;
- displayed $\dim S_2(\Gamma_0(32)) = 1$, displayed $\dim S_2(\Gamma_0(16)) = 0$;
- Sturm pair $32 \cdot \frac{3}{2} = 48$ and $2 \cdot 48/12 = 8$;
- $(\emptyset : \text{Finset } \mathbb{N}).\text{card} = 0$.

The displayed full/new dimensions and the Sturm pair are the LMFDB pin `full=1, new=1, level16=0, sturm=8` for the unique newform 32a1 [6].

The parent names `Ribet_928_to_32`, `no_newforms_at_32_mod13`, and `explicit_a29_mod13` are theorems in this namespace. Each re-exports `no_newforms_at_32`: the numerals of Display 4.1 together with `displayed_frey_matching_newforms.card = 0`. That Finset is defined by `:= ∅`, not computed from the Frey curve’s traces, so its cardinality theorem is immediate and proves nothing about which newform the Frey curve matches. On the parent file `Ribet_Level_Lowering_29_to_32.lean` those three names stay `def Prop`. The theorem named `explicit_a29_mod13` does not compute a_{29} .

Honesty note 4.2 (Newspace dimension 1, not 0). LMFDB records $\dim S_2(\Gamma_0(32))^{\text{new}} = 1$, with unique newform 32a1. The pasted Sage line `CuspForms(32,2).new_subspace().dimension()` is *not* a theorem. The honest Lean display is the empty *asserted* matching Finset (card 0) versus displayed dimension 1; it is a mismatch of definitions, not a derived elimination. Witnesses: `sagemath/ribet_no_newforms_32.sage` and `certs/ribet_no_newforms_32.json` pin `full=1, new=1, level16=0, sturm=8` (`sage_ran: false` in the JSON: these are recorded, not freshly computed, values).

Axioms on the packaged display: `[propext, Quot.sound]`. LLL no-go persistence: `[propext, Classical.choice, Quot.sound]`. No `Lean.ofReduceBool`, no `sorryAx`, no new axiom for either.

A separate, genuinely computed lemma: `no_match_32a1_of_good_red_29`. The file `Level32Table.lean`, imported by `Ribet_Level_Lowering_29_to_32_final.lean` but not used by any of the theorems above, contains a lemma that *is* derived rather than asserted:

$$\begin{aligned} &\text{no_match_32a1_of_good_red_29 } \{A \ B : \mathbb{N}\} \\ &(\text{hsol} : A^4 + B^4 = (B + 3)^{13}) (\text{hA} : 29 \nmid A) (\text{hB} : 29 \nmid B) (\text{hC} : 29 \nmid B + 3) : \\ &\text{frey_ap } 29 (A \bmod 29) (B \bmod 29) \neq (-10 : \mathbb{Z}). \end{aligned} \quad (3)$$

Here `frey_ap p A B` is the trace $a_p = p - \#E_{A,B}(\mathbb{F}_p)$ of the naive reduction of (2) mod p , computed by counting affine points, and -10 is `curve32a1_ap_29`, the a_{29} coefficient of 32a1. The proof reduces (3) to a closed instance, `frey_a29_ne_32a1_fin`: for every pair $(A, B) \in (\mathbb{Z}/29\mathbb{Z})^2$ with $A \neq 0$, $B \neq 0$, $A^4 + B^4 \not\equiv 0 \pmod{29}$, and $A^4 + B^4 \equiv (B + 3)^{13} \pmod{29}$ — i.e. every residue pair consistent with a good-reduction solution of (1) at 29 — the trace is $\neq -10$. This is a finite check over the $29^2 = 841$ candidate pairs, proved by `native_decide`, not `decide`.

Honesty note 4.3 (Axiom profile differs from the four named displays). Rebuilding `Level32Table.lean` on the pinned toolchain and running `#print axioms no_match_32a1_of_good_red_29` reports

`[propext, Classical.choice, Lean.ofReduceBool, Quot.sound]`.

The presence of `Lean.ofReduceBool` is exactly the mark of `native_decide`: this lemma trusts the compiled kernel reduction of a finite check over $29^2 = 841$ residue pairs, unlike every named-display theorem in Sections 2–5, which are stated above to avoid `native_decide` and to have no `Lean.ofReduceBool` in their axiom lists. Both facts are true simultaneously: the four packaged `_final` displays really do avoid `native_decide`, and this separate, uncited lemma really does use it. Citing `no_match_32a1_of_good_red_29` as part of an axiom-clean argument would be inaccurate.

Under its stated hypotheses — a gap-3 solution with good reduction at 29 on all of A , B , and $B + 3$ — the lemma is a real, mechanically checked mod-29 trace mismatch, and the assessment in this draft’s first version understated it. It does not, however, complete an elimination argument for three reasons recorded in Section 1: (i) it says nothing about $29 \mid ABC$; (ii) no Lean theorem shows a hypothetical solution of the gap-3 family must have good reduction at 29, so the hypothesis is not known to hold in any case of interest; and (iii) the identification of a_{29} of the Frey curve’s Galois representation with a_{29} of 32a1 — which is what a completed Ribet/level-lowering argument would supply — is not derived here; `ribet_lowers_Frey_to_level_32` stays an uninhabited `def Prop` (Section 6). The companion theorem `no_sol_with_good_red_29_of_trace_match` makes this last gap explicit: it takes the trace-match hypothesis that Ribet would need to supply as an additional argument and derives `False` from it via `no_match_32a1_of_good_red_29`. Its axiom list, `[propext, Classical.choice, Lean.ofReduceBool, Quot.sound]` (checked the same way), still carries `Lean.ofReduceBool`, since it calls the `native_decide`-based lemma directly rather than merely citing its statement; no theorem in this file removes that dependency.

5. DISPLAYED DATA IV — KOLYVAGIN-STYLE NUMERALS ON 26a1, 26b1

The file `Kolyvagin_MW_Rank0_26a1_26b1_final.lean` (fc8c7a1 / phase merge bc7e93b; incremental 35489926310, 262s) displays numerals for Cremona 26a1 (LMFDB 26.a2) and 26b1 (LMFDB 26.b2) [5, 1, 6].

Display 5.1 (Kolyvagin numerals and typeclass upgrade). • $|\text{Sel}_2|^{\text{disp}} = 1$ (`Sel2_card_one`, `displayed_Sel2_card_26a1/b1`);

- $3 \cdot 7 = 21$ (`three_mul_seven`);
- $L/\Omega = 1/3$ on 26a1, $1/7$ on 26b1 (`norm_num` on $\mathbb{Q}$);
- $\neg \text{IsRankZero}$ on both models;
- $\text{Nonempty}(\text{Fintype } \alpha) \rightarrow \text{Finite } \alpha$ (`finite_iff_nonempty_fintype`);
- $|\text{Unit}| = 1$ (Subsingleton card on `Unit`).

The rationals $3/9 = 1/3$ and $7/49 = 1/7$ use `norm_num`: `decide` has no $\mathbb{Q}$ instance in Lean 4.12.

Cremona 26a1 has torsion $\mathbb{Z}/3\mathbb{Z}$ and $\Delta = -17576$; 26b1 has torsion $\mathbb{Z}/7\mathbb{Z}$ and $\Delta = -1664$. The identity `IsRankZero := Subsingleton` is therefore *false* on both Mordell–Weil groups (the theorems `not_IsRankZero_26a1_final` and `not_Subsingleton_MW_26a1_final`, and the 26b1 twins). The `Subsingleton`→`Fintype` path applies to `Unit`, not to $E(\mathbb{Q})$. The section title above says “numerals,” not “rank 0,” precisely because the repository’s own display file proves `IsRankZero` *fails* for the curves it names.

The parent names `MW_rank_zero_fintype`, `Kolyvagin_L_nonzero_imp_MW_rank_zero`, and `TwoDescent_implies_MW_rank_zero_fintype` are theorems in this namespace. Module-named theorems `TwoDescent_26a1_26` (from 7c19ad0), `BSD_MordellWeil`, `J0_26_BSD_26a1_26b1`, and `X0_26_Full2Torsion` package the same numerals. On the parent files, `MW_rank_zero_fintype`, `Kato`, `BSD_MordellWeil.MW_rank_zero`, and `X0_26_Full2Torsion.J0_26_rank0` stay `def Prop`.

Honesty note 5.2 (Not $E(\mathbb{Q})$ finite). Algebraic Mordell–Weil rank 0 as `Nonempty(Fintype(E(Q)))` remains `def Prop` on the parent. The packaged theorem `MW_rank_zero_fintype` is Display 5.1, not Kato–Kolyvagin and not Sage `E.rank()`. Witness: `sagemath/kolyvagin_rank0_26.sage` with `certs/kolyvagin_rank0_26.json` (`sage_ran: false`: the file records pinned values, not a fresh Sage computation).

Axioms: `[propext, Classical.choice, Quot.sound]`. No `Lean.ofReduceBool`, no `sorryAx`, no new axiom.

6. LEAN FORMALIZATION — EXPLICIT TYPES

Every theorem in the four finals carries an explicit type. Lean 4.12 rejects `theorem foo := bar`. The required opens are `open Nat Finset Classical`. Typical types are $\mathbb{N}$, `Prop`, `Finset  $\mathbb{N}$` , and `Fintype`. Proofs use `decide` (or `norm_num` on $\mathbb{Q}$), never `native_decide`, so `Lean.ofReduceBool` does not appear.

Pin. Toolchain `leanprover/lean4:v4.12.0`. Mathlib commit `809c3fb3b5c8f5d7dace56e200b426187516535a`, inputRev `v4.12.0`. Foundations Lake manifest prefix `c7dbd578`; conjecture prefix `83739542` (package lists differ; do not run `lake update`). Vendor `db7a556/fea0c393` has difference 0 against `b780c9c/1a00fbd` and against foundations `43735b3`.

Axioms. For the four packaged `_final` displays of Sections 2–5, allowed axiom lists are `[propext]`, `[propext, Quot.sound]`, and `[propext, Classical.choice, Quot.sound]`. There is no `sorryAx` and no new axiom, and `Lean.ofReduceBool` does not occur, i.e. none of the four uses `native_decide`. Parent files keep the algebraic names as `def Prop`. Display files contain no new `def Prop`. This does *not* extend to every theorem in every file those four displays import: Section 4 records one lemma, `no_match_32a1_of_good_red_29` in `Level32Table.lean`, whose axiom list is `[propext, Classical.choice, Lean.ofReduceBool, Quot.sound]` because its proof uses `native_decide` on a finite residue check. Both were verified on the pinned toolchain with `lake build +Level32Table` followed by the file’s own `#print axioms` lines (cache warm, ~ 1.3 s), and, for the companion theorem `no_sol_with_good_red_29_of_trace_match`, with a one-line standalone `lake env lean check (import Level32Table; #print axioms no_sol_with_good_red_29_of_trace_match)`, which also reports `Lean.ofReduceBool`.

CI, incremental only. Foundations `build.yml` is `HonestB0Search` incremental: `lake exe cache get`, then `+Tate_Frey_Conductor_29_Neron_final`, `+Mazur_X0_13_No_Isogeny_final`, `+Ribet_Level_Lowering_29_to_32_final`, `+Kolyvagin_MW_Rank0_26a1_26b1_final` (~ 40 s on a warm `.lake`), then `lake build HonestB0Search` and `lake build Level26` (1–2s each). Cache key `lake-${hashFiles(lean-toolchain,lake-manifest.json)}-v27-6ccafbf`. Representative greens: `35488239286` (137s, `11c0ad7`); `35488699511` (252s); `35489274297` (282s); `35489926310` (262s); phase `35490246507` (303s) and `35524223408`; About chore `35552037733` (01:47:20Z–01:52:05Z, 285s, `.lake` restored). On those working branches `main.yml` recorded 0 runs. The phase incremental `35524223408` is the green gate. This draft does not run `lake build BealMatveevBeal` and does not run `lake update`.

Conjecture `main.yml`. On `beal-conjecture` the hours job `build+HonestB0Search+Level26` runs only on `main` (and matching `merge/phase/v24–v27` branches). Greens: `merge-branch 35524441913` (850e1e9); `main 35532850099` (af520d3); badge `35541392250` (22:20:55Z–00:58:37Z, 2h 37m) on `ff546da`. Tip `ea261b9` is the About chore on top of that badge.

7. COMPUTATIONAL VERIFICATION

Zenodo. Version DOI `10.5281/zenodo.22863527` is a new version of `22854603` and `22851334` [3]. The concept DOI `10.5281/zenodo.22379293` stays, recorded as `isVersion0f` [2]. GitHub release `v30.0.0-level-26-structural-verification` has id `392549396` at `af520d3`.

Sage and JSON.. The Lean displays are paired with static Sage scripts and JSON pins, not with a Sage kernel in CI: `sagemath/tate_nero_29.sage`; `sagemath/ribet_no_newforms_32.sage` with `certs/ribet_no_newforms_32.json` (`full=1`, `new=1`, `level16=0`, `sturm=8`); `sagemath/kolyvagin_rank0_26.sage` with `certs/kolyvagin_rank0_26.json`. LMFDB labels used as pins are `32a1`, `26a1`, and `26b1` [6].

Repository surface. Folder READMEs are complete outside the vendor tree (difference 0 against `43735b3`). GitHub About texts are 302 characters (foundations, homepage the concept DOI) and 308 characters (conjecture, homepage the version DOI). Topics include `lean4`, `mathlib`,

beal-conjecture, frey-curve, tate-algorithm, mazur, ribet, kolyvagin, modular-curves, number-theory, mcom (conjecture also x0-13, x0-26). There is no arXiv badge.

Tags and working branches. Foundations tags v0.27, v0.28, v0.29, v0.30 stay live. Phase phase-darmon-merel-4413 stays 2b60ccd (v0.30-mcom-structural-verification). Working finals stay tate-neron-final-v29 11c0ad7, mazur-no-isogeny-final-v29 2f4e16e, ribet-928-to-32-final-v29 5631fd1, kolyvagin-rank0-final-v29 fc8c7a1. Conjecture tag v30.0.0-level-26-structural-verification stays at af520d3; this is the fixed anchor for the Lean material discussed above. This revision itself lives on branch paper-v30-honesty-revision, not yet merged to main at the time of writing; see Section 1 on why self-referential commit hashes for main are avoided.

8. CONCLUSION

This draft records, honestly, what four Lean 4.12 files named after Tate, Mazur, Ribet, and Kolyvagin actually contain for the restricted gap-3 family $A^4 + B^4 = (B + 3)^{13}$: numerals and Finset cardinalities, not proofs of the classical theorems. Tate: $928 = 2^5 \cdot 29$ at $(A, B) = (29, 1)$ only (I_0^*, I_8) , not a Lean Néron conductor for general (A, B) . Mazur: 2184 and cusp card 2 versus genus 0 — no 13-isogeny is excluded. Ribet: $928/29 = 32$ with an *asserted*, not derived, matching Finset $\emptyset$ against LMFDB 32a1 of dimension 1. Kolyvagin: $|\text{Sel}_2| = 1$, $3 \cdot 7 = 21$, $L/\Omega = 1/3$ and $1/7$ on 26a1/26b1, where the parent `IsRankZero := Subsingleton` is mathematically false on both curves. Separately, and more substantively, the repository proves `no_match_32a1_of_good_red_29` (Section 4): a genuine, `native_decide`-checked mod-29 trace mismatch for gap-3 solutions with good reduction at 29, whose hypotheses are not shown to hold anywhere and which is not linked in Lean to Ribet’s level-lowering step. That lemma, not the four asserted displays, is the closest thing in this repository to a mathematically meaningful elimination result, and it is scoped to a single residue and a single restricted family, not to Beal’s conjecture or even to the unrestricted signature $(4, 4, 13)$.

On this basis the draft is not a proof or a partial proof of any instance of Beal’s conjecture. It is a record of what a specific Lean development does and does not establish, offered so that the distinction between a displayed numeral and a mathematical theorem is not lost in translation between the repository and its description. The software versions are v30.0.0-level-26-structural-verification on beal-conjecture (af520d3, About tip ea261b9) and v0.30-mcom-structural-verification on foundations 2b60ccd. Phase stays 2b60ccd. Tags v0.27–v0.30 stay live. Parent algebraic statements remain `def Prop`. No remaining timers.

This draft is a Mathematics of Computation submission, not an arXiv preprint. Cite the version DOI 10.5281/zenodo.22863527 and the concept DOI 10.5281/zenodo.22379293.

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